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A Remarkable 20% Oil Production Increase Through the Integration of Array Production Logging Data Analysis and Acid Stimulation Design and Optimization. A Case Study, Offshore Abu Dhabi

A. Hegazy, A. M. Elbendary, O. Abdulrahman, and E. Negm, Halliburton; A. Serry, K. A. Alhosani, P. Agrawal, A. Najy, and Y. A. Ibrahim, ADNOC

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Abstract

This case study focuses on deploying the multiple array production suite to the maximum safe depth within the horizontal well to achieve comprehensive borehole coverage and capture detailed subsurface production information to understand and evaluate reservoir performance in order to optimize acid stimulation design, allocation, and evaluation, ultimately maximizing oil production and controlling gas production for a better well and reservoir management strategies throughout the lifetime of an oil well.

To facilitate reaching the well's total depth, a friction reducer was employed to assist the deployment of coiled tubing to the maximum total depth required. The methodology employed rigorous planning, equipment selection, and precise field implementation. These measures ensured safe and efficient deployment and acquisition of the multiple array production logging tools to the well's maximum safe depth. This provided detailed flow profile information of the oil and gas entry points across the reservoir to help design the acid stimulation allocation across the oil entry points of the reservoir while maintaining the gas production to a minimum.

The results showcased the impact of deploying and analyzing the array production logging suite on operational efficiency, acid stimulation design and allocation, and overall production performance. The utilization of an array production logging facilitated comprehensive borehole fluid coverage, offering a detailed evaluation of subsurface fluid distribution conditions enabling the identification of oil producing inflow control devices as well as gas producing inflow control devices. The subsequent focused acid stimulation design and allocation through the reservoir section led to 20% increase in oil production while effectively minimizing undesired gas production. These findings highlight the instrumental role of multi-sensor production logging in delivering data acquisition operational efficiency while providing the necessary data to optimize the production outcomes.

The pre-stimulation array production logging acquired data provided a decisive input to support the acid treatment design for the production performance enhancement. The comprehensive borehole coverage provided detailed subsurface fluid distribution evaluation, enabling identification of oil and gas producing

inflow control devices and focused acid stimulation allocation across the oil producing inflow control devices resulting in a remarkable 20% increase in oil production while minimizing gas production.

Introduction

The global implementation of horizontal wells is expanding, owing to their proven effectiveness in enhancing hydrocarbon recovery. By increasing the contact area between the wellbore and the reservoir's productive zones, horizontal wells are designed to maximize production and fulfill targeted well and reservoir management objectives. Offshore Abu Dhabi, the demands for segmented completions in multilateral horizontal drains is growing, especially in mature carbonate fields where the gas and water inflow need to be controlled, while producing thin oil rims without jeopardizing the production targets or the reservoir delivery. There is a coinciding need to evaluate the performance of these complex drains and accurately identify any possible sources of water and gas production. This evaluation is critical for maintaining efficient reservoir management and planning potential intervention strategies.

Research into horizontal flow regimes began several decades ago, driven by the need to understand multiphase flow behavior along horizontal well sections. In response, numerous models have been developed to simulate downhole flow conditions. Among these, the stratified flow regime, where lighter fluid phases flow along the upper portion of the wellbore and heavier phases settle along the bottom, is the most frequently observed under actual reservoir conditions. Phase distribution patterns are significantly influenced by the inclination angle of the well. Wellbores inclined at less than 90 degrees exhibit different phase holdups and flow velocities compared to those inclined at greater than 90 degrees, resulting in distinct flow characteristics. Conventional production logging tools, especially those only employing centerline measurements, often produce inaccurate flow profile data in horizontal wells. This is due to the variable immersion of center-sample sensors across the same horizontal section of the well. Centerline measurements would see water in horizontal sections inclined below 90 degrees, oil in horizontal sections inclined above 90 degrees, or at the interface between the two phases in stratified flow regimes inclined at 90 degrees. These inconsistencies introduce significant uncertainties in the measurement of phase holdups and velocities. As a result, the derived flow profiles may not accurately reflect true downhole conditions, potentially leading to misinformation and poor decisions regarding remedial actions and compromising effective well and reservoir management.

To accurately quantify flow rates in horizontal wells where fluid segregation is anticipated, full-bore array production logging tools have been developed to overcome the limitations of conventional logging technologies. These tools, referred to in this paper as Array Production Logging Tools (APLT), feature multiple sensors positioned at 30–60-degree intervals to cover the entire horizontal cross-section of the wellbore. In addition to full-bore coverage, APLT are equipped with relative bearing measurements to determine the exact orientation of each measurement point within the wellbore. Furthermore, APLT provides measurements of wellbore deviation, enhancing the accuracy and reliability of flow profiling in complex well trajectories.

The APLT system comprises the Spinner Array Tool (SAT), Capacitance Array Tool (CAT), and Resistance Array Tool (RAT), which collectively provide measurements of phase velocities and three-phase holdup fractions at all depths throughout the logging interval along the horizontal drainage section.

The SAT tool is equipped with six miniature turbines mounted on bowspring arms, allowing for cross-sectional velocity profiling. This configuration enables accurate determination of fluid velocities and flow directions throughout the wellbore. The CAT tool features 12 miniature capacitance sensors mounted on bowspring arms, enabling cross-sectional measurement of fluid holdups. This configuration allows for effective differentiation between water, oil, and gas phases along the wellbore, providing three-dimensional imaging of phase distribution. The RAT tool consists of 12 microsensors mounted on bowspring arms, enabling cross-sectional profiling of water holdup. This design allows for accurate differentiation

between water and hydrocarbons along the wellbore, delivering a three-dimensional representation of the water holdup profile. Integrating the three array tools with conventional production logging sensors enables comprehensive flow profiling across the reservoir. This detailed data supports well and reservoir management strategies by accurately identifying water and gas sources, as well as the primary contributing zones, thereby facilitating effective planning and execution of remedial well intervention operations. Historical production data, supported by numerous production logs, have shown that many of these horizontal wells underperform due to uneven reservoir inflow along the wellbore. This non-uniform influx can lead to bypassing of recoverable oil and early breakthrough of water or gas, resulting in increased operational costs associated with fluid handling, treatment, reinjection, and disposal while simultaneously reducing well productivity.

These performance challenges are commonly associated with the segmented completions. To address them, Inflow Control Devices (ICDs) and Autonomous ICDs (AICDs) are installed in many cases to regulate fluid entry along the horizontal drains along with slotted or pre-perforated liners, with the objective of balancing hydrocarbon flow and mitigating early water or gas breakthrough. However, actual downhole conditions may deviate from design expectations. In such cases, Array Production Logging Tools (APLT) provide critical insight into the performance of these completions, enabling evaluation of their effectiveness and supporting the design of appropriate remedial actions, such as targeted stimulation or water/gas shutoff operations.

Well Information and Problem Statement

The subject well was originally drilled and completed in September 1983 as a deviated dual-producer targeting the reservoirs D2 and D3U. A workover in June 2005 converted the well to a single deviated producer completed across the reservoirs C3U, C3M, C3L, and D2. In December 2023, a second workover was executed, during which the well was sidetracked and completed as a dual horizontal producer across the reservoirs B2 and C2–C3.

Figure-1 exhibits the well completion diagram, 4½-inch liner equipped with Autonomous Inflow Control Devices (AICDs) and 11 oil-swellable packers, segmenting the horizontal section into 10 isolated compartments. Post-completion, the well exhibited underperformance, with surface testing indicating high Gas-Oil-Ratio. As a result, an Array Production Logging Tool (APLT) survey was planned to evaluate the flow profile along the horizontal section, identify the gas entry points, and support the design of a targeted acid stimulation program aimed at enhancing well productivity.

- Track-1: Open-hole Gamma Ray
- Track-2: Well Completion Items (packer-1 located at well heel, packer-11 located at well toe)
- Track-3: Well Inclination

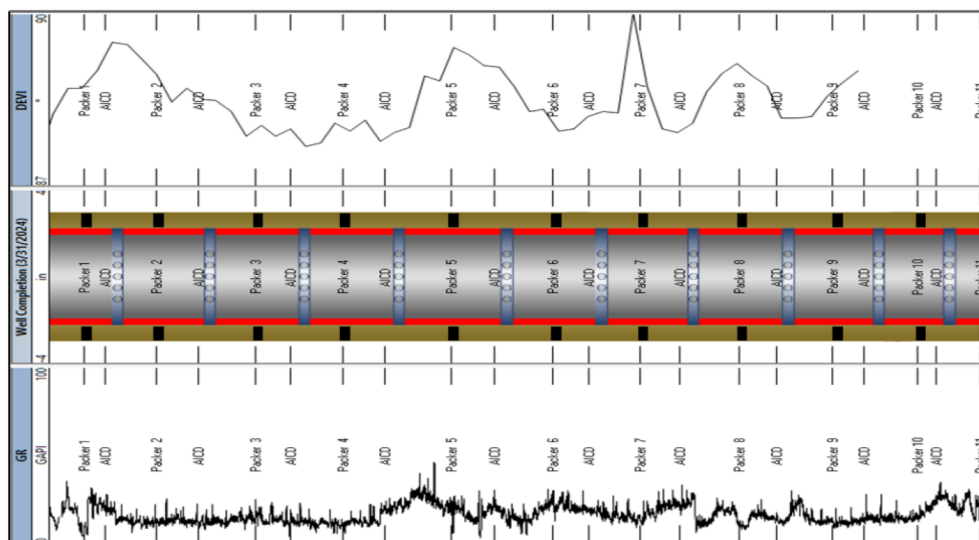


Figure 1—Horizontal Well Completion Diagram

Planning and Deployment Considerations

A structured and comprehensive planning process was undertaken to ensure the success of the production logging operation and any subsequent remedial actions. The following key elements were addressed:

- **Definition of Clear Objectives:** A well-defined job objective was established to guide the planning process and ensure alignment of all operational activities.
- **Review of Well Production History:** Historical production performance was analyzed to better understand reservoir behavior and inform tool selection and logging objectives.
- **Tool Selection and BHA Configuration:** Logging tools were carefully selected to meet the defined objectives. Their placement within the bottomhole assembly (BHA) was optimized to ensure accurate data acquisition and effective tool performance.
- **Deployment Simulation:** Advanced deployment simulations were performed using specialized software. These simulations accounted for deviation surveys, dogleg severity, tool string specifications, tool rigidity, and completion constraints to determine the most suitable deployment method.
- **Logging Program and Operational Design:** A detailed logging program and operational sequence were developed, tailored to the well's specific characteristics and objectives.
- **Risk Assessment and Contingency Planning:** Potential operational risks were assessed, and mitigation strategies were established. Contingency plans were developed to address possible complications such as tool string sticking or failure to reach target depth.

Deployment Method Selection

The method of deployment is a critical factor in determining the overall success of the logging operation. Two primary techniques are commonly utilized: e-line coiled tubing (E-Coil) and wireline tractor deployment. Both methods have demonstrated effectiveness under various operational conditions, with their selection typically influenced by complexity, logistical considerations, and cost-efficiency. A key input in these simulations is the friction factor, which varies significantly depending on wellbore conditions such as open hole versus cased hole and other completion complexities. For E-Coil deployments, simulations predict the maximum depth that can be reached before encountering tubing buckling and estimate the required pulling force and drag forces. In contrast, wireline tractor simulations assess the achievable depth using a

specified cable and verify that the anticipated overpull at total depth (TD) remains within the mechanical limits of the cable.

To manage deployment-related risks, a mitigation strategy was implemented. This included planning for possible tool sticking, restricted reach, or unanticipated wellbore conditions. Mechanically, the production logging tool string was decoupled from the deployment mechanism using a combination of flex joints, knuckle joints, and swivel subs. This design improves tool articulation and minimizes mechanical stress during run-in-hole (RIH) operations.

In this case study, E-Coil was selected as the preferred deployment method based on several operational and risk mitigation considerations:

- The planned post-APLT acid stimulation required precise acid placement across oil-producing compartments, along with the flexibility to perform nitrogen pumping for post-treatment cleanup.
- To address potential deployment challenges, a friction-reducing additive was prepared as part of the risk mitigation strategy, enabling deeper reach during run-in-hole (RIH) in case of early lock-up or high drag conditions.
- Downhole conditions included the presence of free gas with corrosive components (4% H₂S and 5% CO₂) which further supported the selection of E-Coil for safer and more controlled deployment.

Data Acquisition

While multiple passes are standard practice in production logging for vertical wells, this approach is typically impractical in extended-reach horizontal wells due to challenges such as well inclination, completion hardware, deployment-related forces, and elevated friction. Consequently, logging in horizontal wells is generally limited to a single downward and upward pass across the horizontal section. To ensure accurate calibration of the spinner array tool, both flowing and shut-in calibration passes are performed at shallower depths. Figures 2, 3 and 4 present the flowing and shut in spinner calibration passes and plots, where continuous spinners measurements were recorded, enabling reliable computation of spinners slope and threshold values. For array holdup tools, calibration is best conducted in the vertical section of the well, where complete phase segregation can be achieved under shut-in conditions.

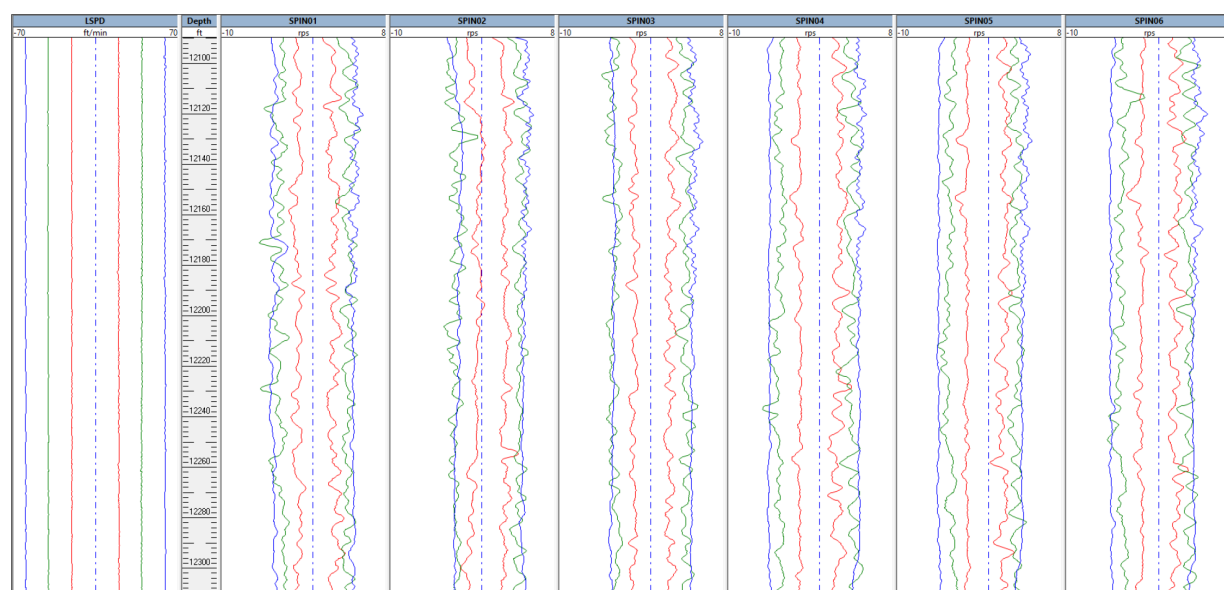


Figure 2—Spinner Array Shut-in Calibration Passes

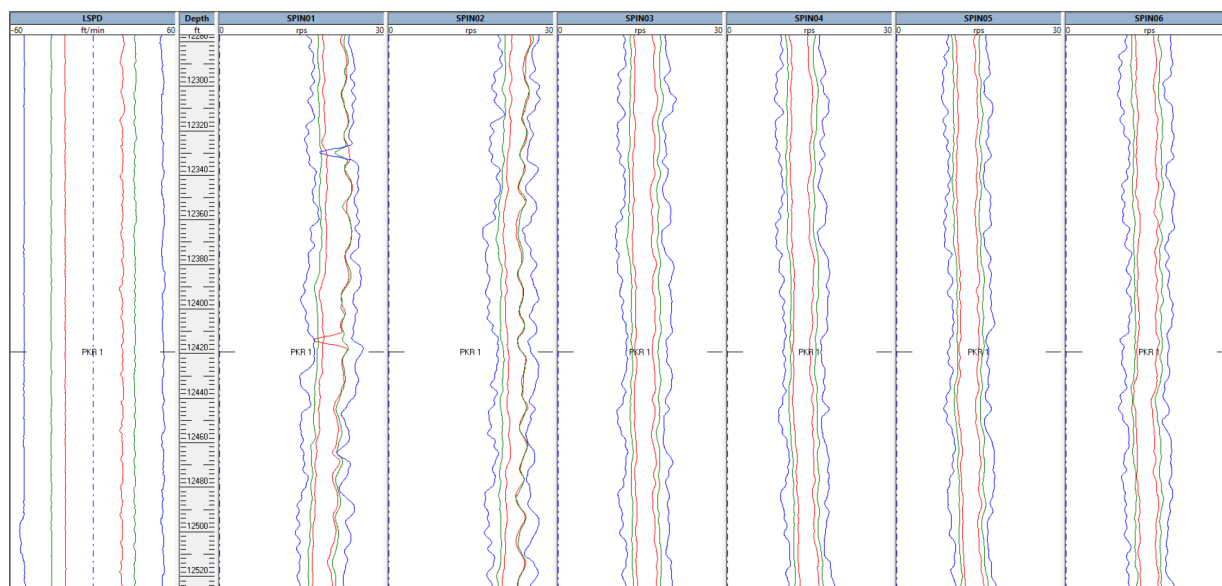


Figure 3—Spinner Array Flowing Calibration Passes

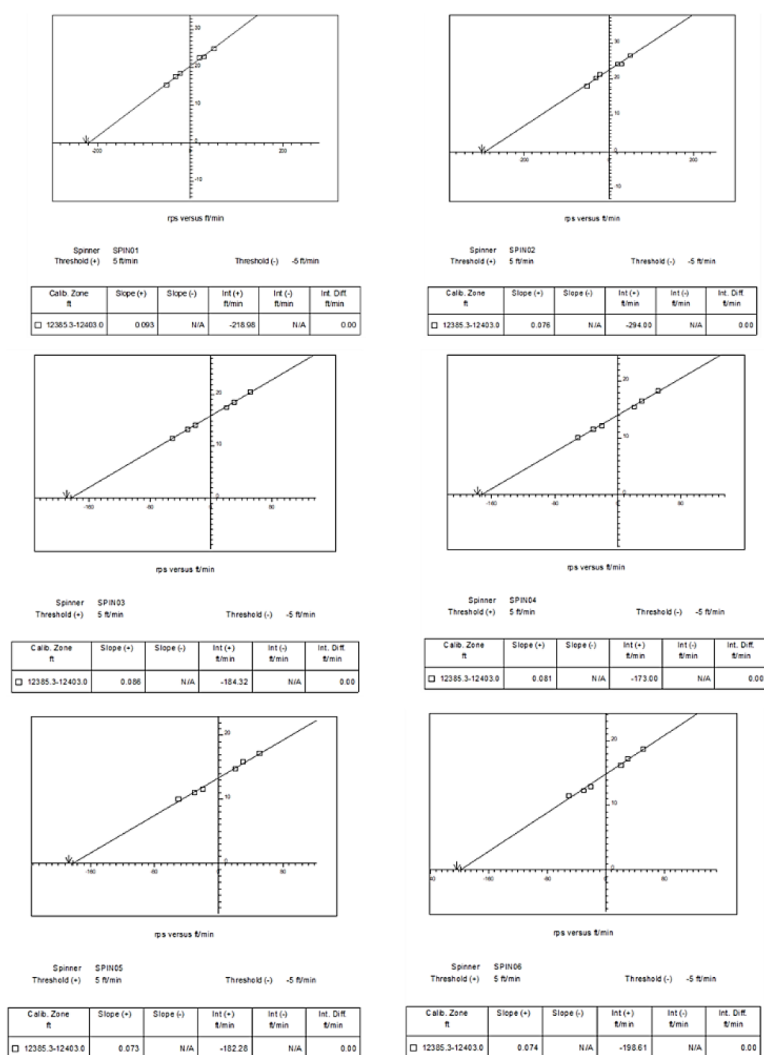


Figure 4—SAT Spinner Calibration Plots In Flowing Condition

The logging operation presented significant challenges in reaching the maximum feasible depth necessary to cover the largest possible portion of the horizontal section in support of acid stimulation planning.

The well was logged in flowing condition at choke 40/64 where the initial downward pass was unable to reach the total well depth due to frictional limitations. To overcome this, a friction reducer was pumped at 0.5 barrel per minute, enabling extended reach into previously inaccessible sections of the horizontal interval. Prior to pumping, the tool reached XX,000 ft approximately 70% of the horizontal section. After pumping the friction reducer, the tool successfully reached YY,000 ft, covering approximately 90% of the horizontal section.

Data Quality Control

The quality control process is initiated during the planning phase, where the appropriate selection of logging tools and deployment methodology is crucial for achieving the defined job objectives. A set of quality control measures is implemented throughout the operation to ensure data integrity and reliability, including:

- Confirmation of both surface and downhole stabilization criteria before initiating main logging passes.
- Accurate depth correlation of acquired data with open-hole gamma ray logs and known completion hardware.
- Consistent and repeatable sensor responses from the array production logging tools (SAT, RAT, and CAT) across the entire horizontal section.
- Continuous and stable velocity readings from the Spinner Array Tool (SAT).
- Pressure and temperature sensors demonstrating consistent behavior across different logging speeds.
- A pressure profile that aligns well with the well trajectory

Figure 5 presents the recorded array production logging data, illustrating consistent, repeatable, and stable responses of the SAT, RAT, and CAT tools. Applied quality control measures validate the integrity of the acquired data, supporting confident interpretation and effective production enhancement planning.

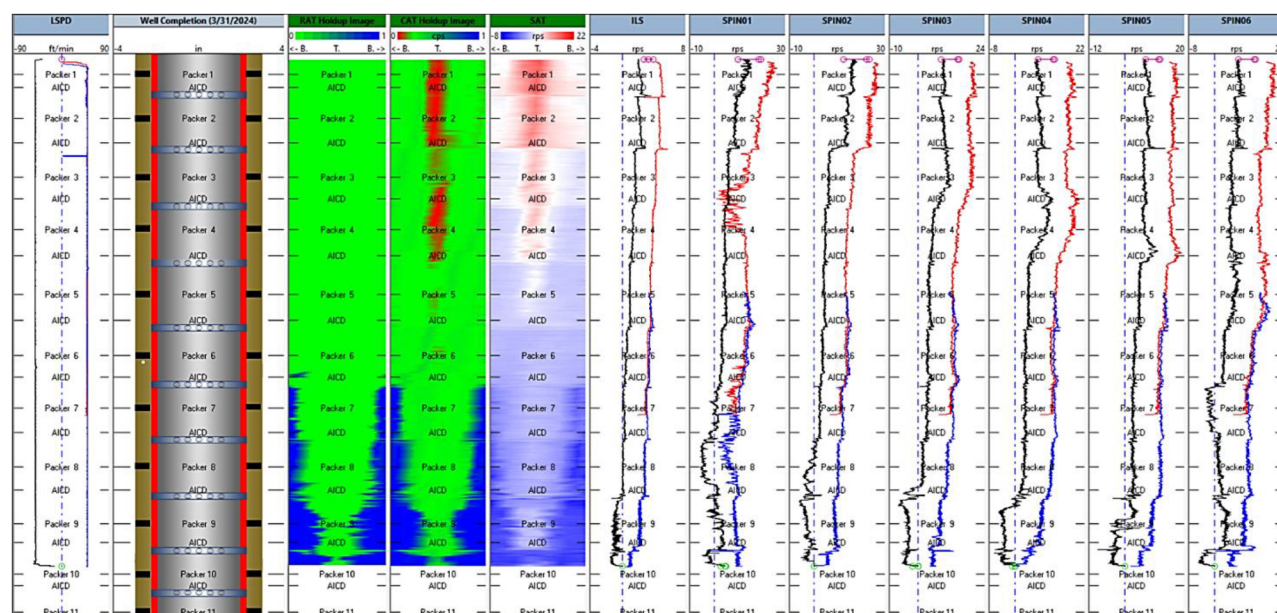


Figure 5—Array Production Logging Recorded Data

The wellbore fluid holdup images from the RAT and CAT tools demonstrate repeatability, revealing stagnant water accumulation near the toe section of the well between packers 6 and 11. Gas inflow is observed in the heel section between packers 1 to 5. The SAT velocity image aligns with the holdup data, showing higher velocities along the upper side of the wellbore, consistent with gas flow in the compartments between packers 1 and 5.

Figures 2 and 3 display the spinner array measurements acquired at varying logging speeds: 20 ft/min (red), 40 ft/min (green), and 60 ft/min (blue).

- Track-1 Logging speeds in feet per minute (ft/min)
- Track-2 Measured Depth in feet (ft)
- Track 3-8 Spinner Array Data Mini-Spinner 1-6

Figure 4 presents the spinner array calibration plots under flowing conditions, showing a well-defined slope that supports the generation of a representative velocity profile.

Figure 5 presents the array production logging data, demonstrating consistent, continuous, and repeatable measurements from the SAT, RAT, and CAT tools.

- Track-1 Logging Speed
- Track-2 Well Completion
- Track-3 Resistance Array Tool Holdup Image (Middle refers to wellbore upper side)
- Track-4 Capacitance Array Tool Holdup Image (Middle refers to wellbore upper side)
- Track-5 Spinner Array Image (Middle refers to wellbore upper side)
- Track-6 Inline Spinner Data
- Track 7-12 Spinner Array Data Mini-Spinner 1-6

Red colored data represents down pass at 50 fpm without pumping friction reducer with maximum depth reached XX,000 ft approximately 70% of the horizontal section. The Blue colored data represents down pass at 50 fpm while pumping friction reducer with maximum depth reached YY,000 ft, covering approximately 90% of the horizontal section. The Black colored data represents up pass from the maximum reachable depth till top log interval.

Array Production Logging Interpretation

The Array Production Logging Tools (APLT) provided continuous circumferential measurements across the horizontal wellbore, enabling detailed characterization of multiphase flow profiles. The Spinner Array Tool (SAT) captures localized velocity data across subdivided cross-sectional layers, while the Capacitance Array Tool (CAT) and Resistance Array Tool (RAT) measurements help for calculation of phase holdup across the same segmented regions. By integrating the velocity data from the SAT with the holdup measurements from the CAT and RAT, a comprehensive flow profile is derived for the entire horizontal section of the wellbore.

Calibration is a critical step in ensuring the accuracy of the array measurements. SAT mini-spinners are calibrated by recording both flowing and shut-in passes in a shallower depth of the well. These calibration runs allow for determination of slope and threshold values essential for producing a reliable velocity profile in the horizontal section. Similarly, CAT and RAT calibration is performed while running in hole under shut-in conditions within the vertical wellbore, where full phase segregation is present. This enables acquisition of reference responses for gas, oil, and water phases, which are then used to interpret data collected in the horizontal section. The APLT interpretation process involves dividing the wellbore cross-section into

discrete segments. Within each segment, average velocity and holdup values are calculated. In a typical three-phase stratified flow regime, where gas occupies the uppermost part of the borehole, oil flows through the middle, and water is concentrated at the lower section, each phase is characterized by its own velocity and holdup distribution along the wellbore. Fundamentally, the interpretation relies on correlating velocity and holdup measurements from sensors aligned at the same circumferential positions. Flow rates are calculated by multiplying the velocity and holdup values in each segment. This segmented approach allows for precise quantification of individual phase contributions along the wellbore. While the basic interpretation model is outlined in the following section, advanced APLT analysis software is employed to integrate all acquired data for a comprehensive evaluation. APLT data interpretation incorporates changes in well inclination into the applied flow models and correlations, ensuring a representative flow profile is generated across the horizontal section.

Basic Interpretation Model of APLT

The fundamental equation for calculating single-phase flow rate is:

$$Q = A \times V$$

- Q is the flow rate,
- A is the cross-sectional area of the wellbore,
- V is the fluid velocity.

For multiphase flow, the calculation incorporates the individual phase velocities and holdups as follows:
The sum of phase holdups is equal to one:

$$Y_g + Y_o + Y_w = 1$$

- Y_g , Y_o , and Y_w represent the gas, oil, and water holdups respectively.

The total flow rate is the sum of the flow rates of each phase:

$$Q_{\text{total}} = Q_g + Q_o + Q_w$$

- Q_g , Q_o , and Q_w represent the gas, oil and water flowrates respectively.

Individual phase flow rates are calculated using the following equations:

$$Q_g = 1.4 \times ID^2 \times Y_g \times V_g$$

$$Q_o = 1.4 \times ID^2 \times Y_o \times V_o$$

$$Q_w = 1.4 \times ID^2 \times Y_w \times V_w$$

- ID: Pipe Internal Diameter, inches
- V_g , V_o , and V_w are the respective velocities of gas, oil, and water
- The factor 1.4 is a unit conversion constant

Figure-6 exhibits the flow profile along the horizontal section, with the toe of the well positioned at packer-11 and the heel at packer-1. The Flow simulation results indicate that a significant gas breakthrough is primarily occurring in the uppermost four compartments near the heel of the well, located between packers 1 and 5. In contrast, the lower compartments between packers 6 and 11, particularly near the toe, shows stagnant water accumulation- without production to surface.

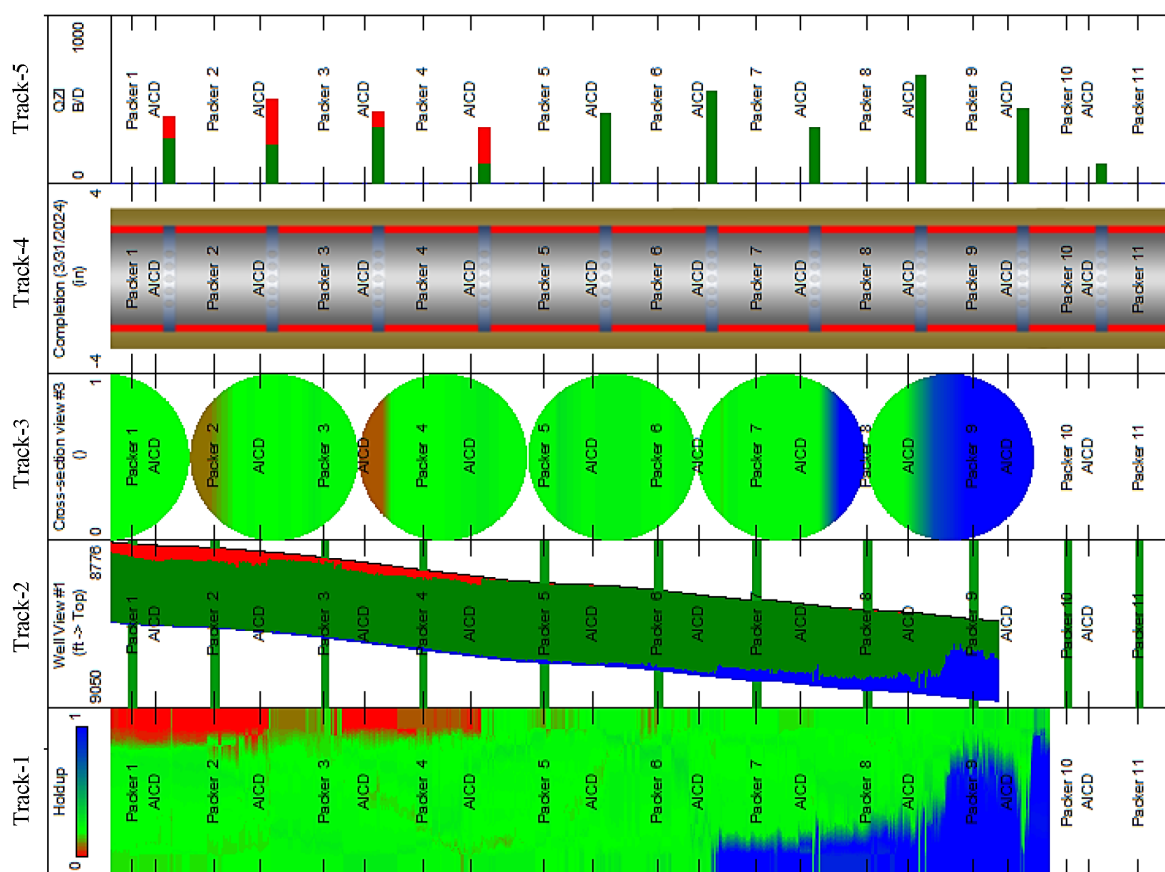


Figure 6—Flow Profile

The observed flow regime is stratified, with gas migrating along the upper side of the wellbore and heavier fluid phases (oil and water) flowing along the lower side. Notably, the compartment between packers 10 and 11 is underperforming, contributing only 2% to the total production.

The interpretation results guided a targeted acid stimulation of the lower drain sections, specifically below Packer-6 in the toe region, due to the identified gas presence near the heel. The treatment was executed using coiled tubing for precise acid placement. Following the stimulation, the well exhibited improved performance, marked by a 20% increase in oil production, a noticeable drop in surface gas-oil ratio (GOR), and a rise in wellhead flowing pressure indicating effective contribution from the stimulated toe compartments.

- Track 1: Wellbore Side View of Holdup Image
- Track 2: Well View
- Track 3: Cross Sectional Holdups
- Track 4: Well Completion
- Track 5: Incremental zonal contribution from each AICD compartment

Conclusion

This case study demonstrates the impact of array production logging (APLT) in guiding effective acid stimulation for horizontal wells in mature fields. By providing detailed insight into inflow distribution and identifying the underperforming zones, the logging data enabled a targeted stimulation strategy aligned with the well and reservoir management objectives, leading to enhancing the well performance. The findings

highlight the value of combining reservoir, petrophysical, and stimulation engineering with advanced production diagnostics to enhance stimulation effectiveness and maximize well productivity.

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Abbreviations

BPD	Barrels Per Day
MMscf/d	Million Standard Cubic Feet Per Day
GOR	Gas-Oil Ratio
WC	Water cut
RIH	Run in Hole
POOH	Pull Out of Hole
PLT	Production Logging Tool
APLT	Array Production Logging Tool
MAPS	Multiple Array Production Suite
CAT	Capacitance Array Tool
RAT	Resistance Array Tool
SAT	Spinner Array Tool
AICD	Autonomous Inflow Control Device
CTU	Coiled Tubing
GR	Gamma Ray
CCL	Casing Collar Locator
FBHP	Flowing Bottom-hole Pressure
SBHP	Shut in Bottom-hole Pressure
BHA	Bottom-hole Assembly
LSPD	Logging Speed
E-Coil	Electric-line Coiled Tubing
STB/d	Stock Tank Barrel Per Day

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